

Classes of Chemical Reactions

Reaction Type	General Form and Example
Precipitation	$WX(aq) + YZ(aq) \longrightarrow WZ(s) + YX(aq)$ e.g. $AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$
Acid-Base	$Acid(aq) + Base(aq) \longrightarrow Salt(aq) + H_2O(l)$ e.g. $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$
Redox	Electrons transfer from the reductant to the oxidant. e.g. $Zn(s) + CuCl_2(aq) \longrightarrow ZnCl_2(aq) + Cu(s)$
Combustion (Hydrocarbon)	$C_xH_y + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$ e.g. $CH_4(g) + 2 O_2(g) \longrightarrow CO_2(g) + 2 H_2O(g)$
Synthesis	$X + Y \longrightarrow XY$ e.g. $2 H_2(g) + O_2(g) \longrightarrow 2 H_2O(l)$
Decomposition	$XY \longrightarrow X + Y$ e.g. $2 H_2O_2(aq) \longrightarrow 2 H_2O(l) + O_2(g)$
Single-Displacement	$X + YZ \longrightarrow XZ + Y$ e.g. $Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$
Double-Displacement	$WX + YZ \longrightarrow WZ + YX$ e.g. $BaCl_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2 NaCl(aq)$
Specialized Reactions	<i>Note: Metal (M), Acid Anhydride (AA), & Base Anhydride (BA)</i>
Metal + Acid	$M(s) + HX(aq) \longrightarrow MX(aq) + H_2(g)$ e.g. $2 Na(s) + 2 HCl(aq) \longrightarrow 2 NaCl(aq) + H_2(g)$
Metal + Water	$M(s) + H_2O(l) \longrightarrow MOH(aq) + H_2(g)$ e.g. $2 Na(s) + 2 H_2O(l) \longrightarrow 2 NaOH(aq) + H_2(g)$
Metal + CO₂	$M(s) + CO_2(g) \longrightarrow MO(s) + C(s) \quad (M_{Reactivity} \geq Al_{Reactivity})$ e.g. $(s) + CO_2(g) \longrightarrow MO(s) + C(s)$
Acid Anhydride + Water (AKA Nonmetal Oxide + Water)	$AA + H_2O(l) \longrightarrow HX(aq)$ e.g. $SO_2(g) + H_2O(l) \longrightarrow H_2SO_3(aq)$
Acid Anhydride + Base (AKA Nonmetal Oxide + Base)	$AA + MOH(aq) \longrightarrow Salt(aq) + H_2O(l)$ e.g. $SO_2(g) + 2 NaOH(aq) \longrightarrow Na_2SO_3(aq) + H_2O(l)$
Base Anhydride + Water (AKA Metal Oxide + Water)	$BA + H_2O(l) \longrightarrow MOH(aq)$ e.g. $CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq)$
Base Anhydride + Acid (AKA Metal Oxide + Acid)	$BA + Acid(aq) \longrightarrow MX(aq) + H_2O(l)$ e.g. $MgO(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2O(l)$